F Training Details

This section outlines the detailed configurations of the datasets and models used in our experiments, including dataset splits, pre-processing steps, model architectures, and training settings.

F.1 Datasets

Below we discuss the 6 datasets used in our paper.

Emotions dataset, proposed in Saravia et al. [2018], consists of 16,000 train, 2,000 validation and 2,000 test samples, each sample belonging to one of the 6 classes (class 0-5): sadness, joy, love, anger, fear and surprise. To induce the notion of noisy labels, we randomly flip labels of 1% of class 5 train samples to any another random class label and let the model train till we reach 100% train accuracy (memorizing the noisy labels).

TweetTopic dataset, developed in Antypas et al. [2022], consists of 2,858 train, 352 validation, and 376 test samples, spanning across 6 classes (class 0-5): arts_&_culture, business_&_entrepreneurs, pop_culture, science_&_technology, sports_&_gaming, daily_life. To introduce memorization of noisy labels, we flip labels of 1% of class 3 train samples to any another random class label while training the model till it reaches 100% train accuracy.

News dataset, proposed in Okite97 [2024], consists of 4,686 train and 828 test samples, spanning across 6 classes (class 0-5): business, sports, politics, health, entertainment and tech. We then split the train set to train & validation using 90:10 stratified split over the class labels. Also, we introduce noisy labels, by flipping labels of 1% of class 5 train samples to any another random class label, while letting the model overfit to 100% train accuracy.

CIFAR10 dataset, first introduced in Krizhevsky et al. [2009], consists of 60,000 samples, with each belonging to one of the 10 classes (class 0-9): airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. In our setup we consider a subset of - 16,000 train, 4,000 validation and 10,000 test samples. We also resize the images to size (224x224x3) for training the transformer model. To introduce noisy labels, we flip labels of 1% of class 9 train samples to any another random class label, and train the model till it reaches 100% train accuracy.

UTK-Face dataset, proposed in Zhang et al. [2017], consists of 23,705 samples and 5 classes (classs 0-4) depicting ethnicity: white, black, asian, indian and others. We split the dataset into train, validation and testing using 65:15:20 stratified split. We also resize the images to size (224x224x3) for training the transformer model. We then introduce noisy labels, by flipping labels of 1% of class 2 train samples to any another random class label, and train the model till it achieves 100% train accuracy.

NICO++ dataset [Zhang et al., 2023] consists of approximately 60,000 images distributed across 60 categories of everyday objects. In this study, we select a subset of 15 object categories, including car, flower, penguin, camel, chair, monitor, truck, wheat, sword, seal, lion, fish, dolphin, lifeboat, and tank. The dataset is partitioned into training (80%), validation (10%), and testing (10%) sets using stratified sampling. Images are resized to (224x224x3) for consistency with the model input requirements. We then introduce noisy labels, by flipping labels of 1% of class 6 train samples to any another random class label, and train the model till it achieves 100% train accuracy.

F.2 Models

Below we discuss the 13 transformer models (6 Post-LN and 7 Pre-LN) considered as part of our paper. We utilize the Sequence Classification variant of all the models available on Huggingface[3].

[3] `https://huggingface.co/docs/transformers/index`

Post-LN Models	Description
BERT [Devlin et al., 2019]	12-layer bidirectional transformer for masked language modeling and next sentence prediction.
DeBERTa [He et al., 2020]	12-layer model with disentangled position/content embeddings and decoding-enhanced attention.
RoBERTa [Yinhan et al., 2019]	12-layer robustly optimized BERT variant trained with dynamic masking and more data.
ELECTRA [Clark, 2020]	12-layer model using replaced token detection for sample-efficient pre-training.
DistillBERT [Sanh et al., 2019]	6-layer distilled BERT that is smaller and faster variant of BERT, retaining strong performance.
Longformer [Beltagy et al., 2020]	12-layer model with sparse attention for efficient long-sequence processing.
Pre-LN Models	**Description**
GPT2-Medium [Radford et al., 2019]	24-layer unidirectional transformer trained for causal language modeling.
GPTNeo-125M [Black et al., 2022]	12-layer open-source causal language model trained on The Pile dataset
Qwen2-0.5B [Yang et al., 2024]	24-layer efficient LLM optimized for generative tasks, using RMSNorm [Zhang and Sennrich, 2019]
RoBERTA-PreLayerNorm [Ott et al., 2019]	24-layer variant of RoBERTa with Pre-LN setting for improved training stability.
ViT-B [Alexey, 2020]	12-layer Vision Transformer Base model for image classification.
ViT-S [Assran et al., 2022]	12-layer smaller ViT variant trained with Masked Siamese Networks (MSN).
DeiT [Touvron et al., 2021]	12-layer Data-efficient Image Transformer trained with distillation, without external data.

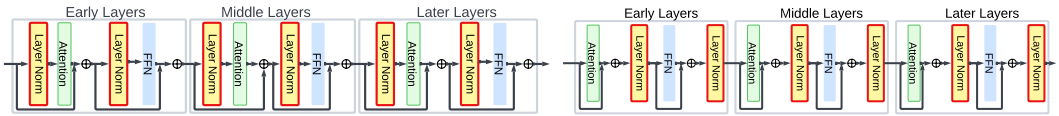
Table 3: Overview of the 13 transformer models categorized into 6 Post-LN and 7 Pre-LN architectures.

F.3 Training Settings & Hyperparameters

In our study, we explore various combinations of different models and datasets across our experiments, as follows - (1) Emotions dataset with BERT (Post-LN), DeBERTa (Post-LN), DistillBERT (Post-LN), GPTNeo (Pre-LN), (2) News dataset with ELECTRA (Post-LN), Longformer (Post-LN), Qwen2 (Pre-LN), (3) Tweets dataset with RoBERTa (Post-LN), GPT2 (Pre-LN), RoBERTa-PreLayerNorm (Pre-LN), (4) CIFAR10 dataset with ViT-B (Pre-LN), (5) UTK-Face dataset with DeiT (Pre-LN), and (6) NICO++ dataset with ViT-S (Pre-LN). We fully unfreeze all layers of the pre-trained models during training.

Regarding the hyperparameters, we consider a learning rate of 2e-5 and a batch size of 16 for all models. Then, we train the Post-LN models for 40 epochs and Pre-LN models for 70 epochs. We run 70 epochs for Pre-LN models, because after removal of LN parameters, the learning accuracy is impacted significantly in Pre-LN models from the start. Hence, we examine if the accuracy would increase by letting the model train more. In addition to that, we do not use any data augmentation in our training procedures. We also provide the code for our experiments in the supplementary file.

F.4 Grouping of Early, Middle, and Later Layers



(a) Pre-LN architecture (b) Post-LN architecture

Figure 5: Pre-LN vs. Post-LN architectures depicting LN placement and categorization of early, middle and later layers

To investigate which group of layers' Layer Normalization (LN) contributes most to memorization and learning, we divide the transformer layers into three subsets: **Early**, **Middle**, and **Later** layers as shown in Fig. 5.

Formally, for a transformer model with N layers (where N is divisible by 3), we define:

$$\begin{aligned} \text{Early Layers} &= \{1, 2, \ldots, \tfrac{N}{3}\} \\ \text{Middle Layers} &= \{\tfrac{N}{3} + 1, \ldots, \tfrac{2N}{3}\} \\ \text{Later Layers} &= \{\tfrac{2N}{3} + 1, \ldots, N\} \end{aligned} \tag{112}$$

This grouping helps separately examine which set of layers most significantly influences learning and memorization in transformers.

G Additional Experiments & Results

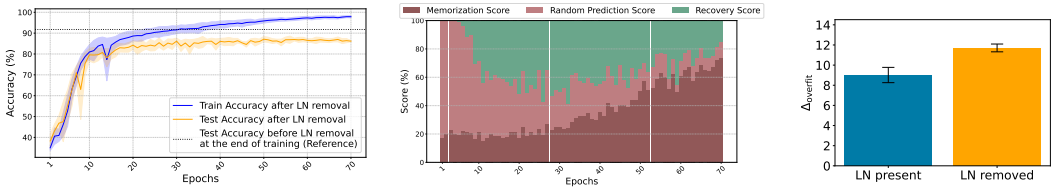
This section presents additional experiments (on top of the ones discussed in Sec. 4, Sec. 5 and Sec. 6)and results on supplementary datasets and models, expanding on the analyses in Sec. G.1, G.2, and G.3. These experiments aim to: (1) establish the distinctive impact of Layer Normalization (LN) on memorization and learning in Pre-LN vs. Post-LN models, (2) assess the role of LN in early layers, and (3) investigate how gradient behavior accounts for the observed phenomena.

G.1 Impact of LN on Memorization & Learning in Pre- and Post-LN models

In this section, we present the results corresponding to the distinctive impact of LN of memorization and learning for the remaining Pre and Post LN models spanning multiple datasets. These results further corroborate our finding that removal of LN parameters in pre-LN models critically destabilizes learning while exacerbating overfitting, while for Post-LN models, LN parameters removal, suppresses memorization and facilitates true label recovery without impacting learning.

The results are verified against Pre-LN models - GPTNeo, GPT2, ViT-B, DeiT, ViT-S, RoBERTa-PreLayernorm and Post-LN models - BERT, DeBERTa, Longformer, RoBERTa, DistilBERT spanning across multiple language and vision datasets - Emotions, News, Tweets, CIFAR10, NICO++, UTK-Face, are provided in Figs. 6, 7, 8, 9, 10, 11, 12, 13, 14,15. and 16.

G.1.1 Pre-LN models - Learning Destabilized



(a) Learning (Test) accuracy over epochs for Pre-LN Model (GPTNeo)

(b) Memorization, Recovery and Random Predictions over epochs for Pre-LN Model (GPTNeo)

(c) Overfitting gap for Pre-LN Model (GPTNeo)

Figure 6: **LN removal destabilizes learning in Pre-LN model - GPTNeo, Emotions Dataset:** LN removal critically affects learning while memorization still persists in GPTNeo. This further exacerbates overfitting, explained by increasing train-test accuracy gap when LN is removed, due to the drop in test-accuracy.